

Differentiable pull-in solver

solves the saddle-node bifurcation directly · exact gradients, no surrogate fitting · no training data

Inverse design

22.86 → 13.60 V

–41% drive voltage, electrode shape alone

15 s on a laptop CPU

no new mask, no commercial licence

Amortized network

0.04 ms per design

spec → geometry, 0 stored simulations

0.56% spec error

hard-constraint layer, not a penalty

Safe-operation RL

+12.8% usable travel

0.0% of devices destroyed

learns each device's ceiling

which cannot be measured